

Data-Driven Dynamics Modeling of a 9-Degree-of-Freedom Rehabilitation Robot Based on the Koopman Operator

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Abstract—A study has been conducted on the dynamics of a serial-parallel hybrid redundant actuation rehabilitation robot. Compared to traditional robots, the dynamic modeling of redundant robots presents greater challenges. The number of actuators exceeds the minimum degrees of freedom (DOFs) required to complete the task, leading to multiple solutions for task planning. As the number of DOFs increases, so does the number of state variables in the dynamic equations, with the coupling effects between joints resulting in a highly nonlinear relationship among joint forces, velocities, and accelerations. Consequently, deriving a dynamic mechanistic model based on Lagrange or Newton-Euler equations becomes increasingly complex. In recent years, Koopman operator theory has attracted growing attention in the modeling of nonlinear dynamic systems. The core idea behind this theory is to lift nonlinear systems into a high-dimensional space, where their evolution can be described by linear operators. Extended Dynamic Mode Decomposition (EDMD) is a tool that facilitates the implementation of linear dimensionality expansion and the modeling of system dynamics. This study adopts the EDMD-based Koopman operator method to establish the dynamic model of the redundant robot, transforming nonlinear problems into linear ones in high-dimensional space. This approach circumvents the complex derivation and solution of nonlinear equations, thereby simplifying model construction. The resulting dynamic model demonstrates high predictive accuracy.

Keywords—Rehabilitation Robot; Redundant Drive; Dynamic Modeling; Koopman Operator.

I. INTRODUCTION

The robot dynamics model primarily studies the motion and mechanical behavior of robots, including joint motion, the calculation of forces and torques, the system's dynamic response,

and its interaction with the external environment. These studies help in understanding and predicting the performance of robots during task execution, thereby optimizing control strategies and designs. The dynamic model enables precise motion control and trajectory planning, prediction of joint forces and torques, optimization of design, fault detection, and diagnosis, and is crucial for the development and control of robots.

Common dynamic modeling methods for robots include the Lagrange method, the Newton-Euler method, and the recursive method. The Lagrange method establishes a dynamic model based on the kinetic and potential energy of a system, making it suitable for dealing with complex multi-degree-of-freedom (DOF) systems and capable of effectively handling constraints and nonlinearity. The Newton-Euler method, grounded in Newton's and Euler's laws, derives dynamic equations through joint-by-joint analysis. This method is suitable for real-time control and motion planning, and it is commonly used in industrial robots. The recursive method calculates the force and motion of joints through a hierarchical structure, offering high efficiency and real-time performance, making it suitable for dynamic environments [1].

Establishing a dynamic mechanism model for redundant robots is particularly challenging, especially when dealing with multi-DOF or nonlinear systems. As the number of DOFs increases, the dimension of the state space grows. When deriving the Jacobian matrix, it is necessary to consider the impact of each joint on the end effector, which increases computational complexity. The motion coupling between joints can further complicate the dynamic equations. The movement of a specific joint may affect the torque requirements of other joints,

necessitating an accurate description of these coupling relationships in the model, which requires complex matrix operations. The dynamic model of redundant robots often involves nonlinear equations. These nonlinearities arise from the geometric relationships of robot joints, elastic elements, and frictional forces. During model derivation, nonlinear terms can make the solution more complex, making it difficult to obtain analytical solutions; hence, numerical methods are typically required for approximation. In dynamic environments, robots need to respond rapidly, which requires that the dynamic model's solution be completed in real-time. The complex calculations induced by redundant structures may lead to response time delays [2, 3].

The data-driven modeling method for robot dynamics offers significant advantages. It demonstrates strong adaptability and can be updated in real-time to accommodate task changes, making it suitable for a variety of applications. Additionally, this method effectively handles complex nonlinear dynamic behaviors, especially exhibiting superior performance in high-dimensional data scenarios. Compared to traditional mechanism modeling methods, it significantly reduces the computational cost of modeling and improves model accuracy through extensive data training, thereby avoiding the uncertainty errors inherent in traditional models. Furthermore, data-driven methods support the autonomous learning of robots, enabling continuous performance optimization through reinforcement learning and imitation learning. These methods possess strong universality and transferability, making them applicable across various tasks and platforms, which further enhances development efficiency and safety. Therefore, this approach is driving the rapid advancement of robotics technology. Common data-driven modeling methods include supervised learning, deep learning, system identification, and reinforcement learning. Compared to methods such as neural networks, which require large amounts of data and entail high computational complexity, the EDMD-based Koopman operator theory offers a more efficient data-driven modeling approach. This method enables the construction of globally linearized dynamic models of robots, simplifying the dynamics modeling process, achieving accurate predictive performance, and enhancing control efficiency[4, 5].

In order to solve the above problems, this article conducts research on a redundant data-driven modeling method for rehabilitation robot dynamics. The following models and methods section include: **A.** Introduction of the redundant rehabilitation robot; **B.** Principle of motion decoupling allocation; **C.** Koopman dynamics modeling based on EDMD.

II. MODELS AND METHODS

A. Introduction of the redundant rehabilitation robot

The robot studied in this article is a lower-limb end rehabilitation robot with a series-parallel hybrid configuration, which has 9 degrees-of-motion(DOMs). The end effector is a motion pedal that can perform humanoid gait movements to assist patients with balance disorders in moving along predefined trajectories, thereby achieving rehabilitation training. Two sets of 9 DOMs devices are symmetrically arranged to provide support and motion assistance to the patient's lower limbs. The robot adopts a modular design in terms of structure, consisting of a 6 DOF parallel device and a 3DOF parallel

device with embedded connections. As a basic 6 device, the 6 DOF device has high stiffness and strong load-bearing capacity. As an extension device, the 3 DOF device has a compact structure and flexible motion, which can increase the working space, motion flexibility, and operability of the robot. By decoupling motion allocation, the load distribution on the 6DOF device can be improved, enhancing the mechanical characteristics of the robot. **Figure 1** shows the robot [6, 7].

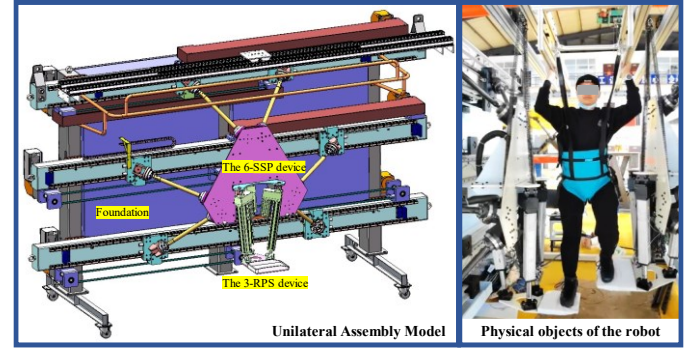


Figure 1. The structure of the robot

The 6 DOF robot consists of 6 spherical joint-spherical joint-prismatic joint (SSP) motion chains, while the 3 DOF robot is composed of 3 revolute joint-prismatic joint-spherical joint (RPS) motion chains. During operation, the two devices must work in coordination and complement each other to achieve the humanoid lower limb gait motion. To improve the effectiveness of rehabilitation therapy, the rehabilitation robot needs to have sufficient workspace and excellent mechanical properties, allowing it to provide patients with a more realistic motion experience.

B. Decoupled Motion Distribution Principle

Redundant robots need to allocate the DOFs of two modular devices effectively, ensuring that they can complete rehabilitation training tasks while minimizing unnecessary coupling between joints, thus ensuring both precision and movement efficiency. The core of decoupled motion distribution lies in the independence of the robot module's movements, minimizing the impact of each joint's motion on others, thereby achieving coordinated control.

The configuration of the DOFs for this rehabilitation robot must enable it to perform specific rehabilitation training tasks, such as simulating humanoid lower limb gait movements and assisting patients with balance impairments in completing predefined rehabilitation trajectories. For walking on flat ground, the robot needs to simulate the translation of the foot along with toe flexion and dorsiflexion movements. The mobility of this redundant robot far exceeds the DOFs required for the task; therefore, under the premise of ensuring task completion, the excess DOFs can be allocated reasonably to enhance patient comfort, improve the robot's dynamic characteristics, and reduce control complexity. The range of motion for the joints must not exceed the physiological limits of the human body, and movement speed should be controlled to prevent secondary injuries to the patient. When allocating DOFs, it is essential to coordinate the movements of the robot's various module devices, avoiding significant coupling while ensuring that the

distribution of DOFs aligns with the movement advantages of the modular devices. This optimizes load distribution during the working process and enhances mechanical performance. Additionally, the complexity of the control system must be taken into account, simplifying the decoupling strategy to improve system response speed and control accuracy.

In the rehabilitation robot, the 6SSP device's transmission mechanism consists of a rail and slider, which provides high transmission efficiency during the linear motion of the end effector. However, when the end effector rotates, the movement and power from the power source need to be transmitted through a combination of linear joints, spherical joints, linkages, and spherical joints. The multiple joints and connecting parts increase the length of the transmission chain due to friction and wear, resulting in increased energy loss and reduced effective transmission efficiency. This device features six closed kinematic chains that influence each other and exhibit coupling effects, making precise control and positioning of posture angles challenging. In contrast, the 3RPS device's compact mechanism, with fewer joints and a smaller overall size, allows multiple rotational joints to quickly adapt to different rotational paths, facilitating easy calculation of control posture changes, thus offering flexibility and efficiency in executing rotation tasks.

When allocating DOFs, the structural characteristics of the robotic device must be fully considered. The 6SSP device facilitates linear motion, while the 3RPS device enables rotational motion. The rationality of each joint's range of motion and constraints must be carefully considered to ensure that the allocation of DOFs does not lead to mechanical conflicts or restricted movement. The allocation of DOFs should align with the structural characteristics and performance of the modules, ensuring coordinated movement among the various modules, thereby completing complex tasks more efficiently. This guarantees the uniqueness of the robot's kinematic solutions, reduces the complexity of control algorithms, and enhances the real-time performance and accuracy of the control system. The definition of the robot's coordinate system and the allocation of DOFs are illustrated in Figure 2 below.

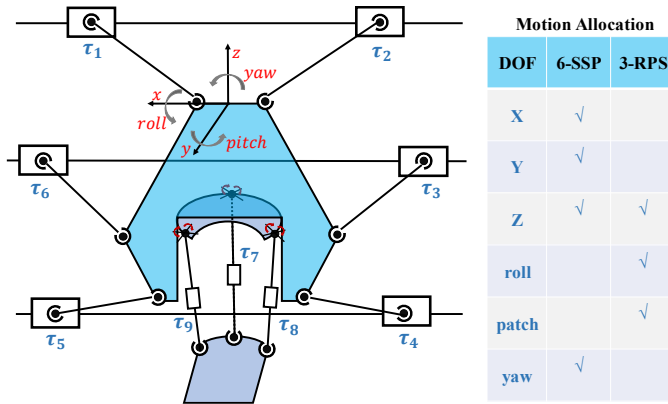


Figure 2. The definition of the robot's coordinate system and the allocation of DOF

III. DYNAMIC MODELING BASED ON KOOPMAN OPERATOR

The Koopman operator method is also one of the data-driven modeling approaches. The core idea of the robot dynamics

modeling method based on the Koopman operator is to map nonlinear dynamic systems to high-dimensional linear spaces, thereby utilizing linear system theory to analyze and control complex dynamics. The advantage of this method lies in its ability to handle complex nonlinear behaviors while capturing the global characteristics of the system, thus improving prediction accuracy and control performance. Compared to models like deep learning, the robot dynamics modeling based on the Koopman operator has the advantage of rigorous theoretical foundations and interpretability. The Koopman operator, by mapping nonlinear systems to high-dimensional linear spaces, can provide clear dynamic characteristic analysis, facilitating the understanding and control of system behavior. In contrast, neural network models are often black-box in nature, making it difficult to interpret their internal mechanisms. Furthermore, the Koopman method can effectively capture the global dynamic features of the system without being significantly affected by data noise. This makes the modeling based on the Koopman operator more reliable and predictable in many practical applications [8, 9].

For a linear time-invariant system, the evolution of its state variables satisfies the Equation 1:

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) \quad (1)$$

In the equation, $\mathbf{x}_k \in \mathbf{R}^n$ represents the state variable at the k -th time step, satisfying the constraint $\mathbf{x}_k \in \mathbf{X}$. $\mathbf{u}_k \in \mathbf{R}^m$ represents the control variable at the k -th time step, satisfying the constraint $\mathbf{u}_k \in \mathbf{U}$. $\mathbf{f} : \mathbf{R}^n \times \mathbf{R}^m \rightarrow \mathbf{R}^n$ is a continuous function with respect to $\mathbf{x}_k$ and $\mathbf{u}_k$.

Assuming that $\mathbf{x}_k$ observable, the Koopman operator is defined as an infinite-dimensional linear operator $\mathbf{K}$ acting on the observation function $\mathbf{g}(\mathbf{x})$. By using $\mathbf{g}(\mathbf{x})$, the observed values of the state variable can be obtained, and then through the operator $\mathbf{K}$, the original system space is transformed into an infinite-dimensional space $\mathbf{K}$, where the state evolves linearly through $\mathbf{K}$. The operator $\mathbf{K}$ cannot be directly applied to the actual space; it requires dimensionality reduction to seek a finite-dimensional matrix $\mathbf{K}$ to approximate $\mathbf{K}$. This study uses the extended dynamic decomposition method to approximate the Koopman operator.

The characteristic function of the Koopman operator is defined as $\phi_k(\mathbf{x})$, and the corresponding eigenvalue is λ_k . The relationship is shown in Equation 2

$$\mathbf{K} \cdot \phi_k(\mathbf{x}) = \phi_k(\mathbf{f}(\mathbf{x})) = \lambda_k \cdot \phi_k(\mathbf{x}) \quad (2)$$

Using the characteristic functions of the Koopman operator $\mathbf{K}$ as basis functions, the observation function can be expressed in this space as Equation 3:

$$\mathbf{g}(\mathbf{x}) = \sum_k \phi_k(\mathbf{x}) \cdot \mathbf{m}_k \quad (3)$$

Therefore, select the first- n eigenvalues corresponding to their characteristic functions (dominant modes). These characteristic functions can represent the vast majority of the features of the Koopman operator and elevate the original space

to an n -dimensional space. These functions constitute the state lifting functions $\{\varphi_1(x), \varphi_2(x) \dots \varphi_n(x)\}$ [10, 11]. The basic idea of the Koopman operator based on EDMD is illustrated in the **Figure 3** below.

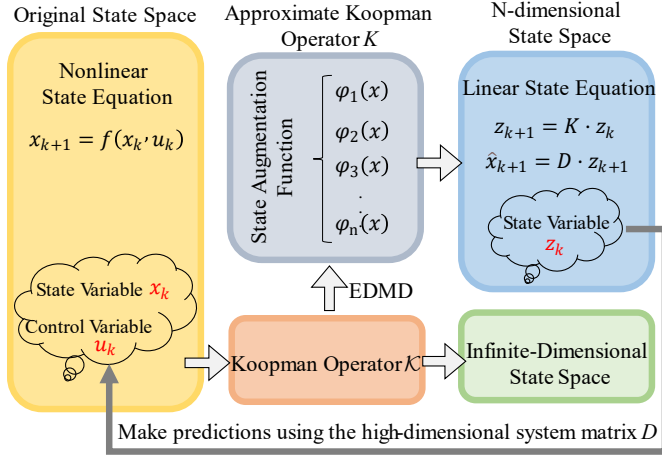


Figure 3. The basic idea of the Koopman operator based on EDMD.

The Koopman dynamic modeling process is as follows:

(1) Data Acquisition and Preprocessing

This study establishes the relationship between the robot's motion state and control torque through data-driven methods, requiring a large amount of prior data to learn model features. The dynamics simulation involves ensuring that the robot's motion is as widely distributed as possible within the workspace. According to operational requirements, the robot simulates humanoid gait movements, working along predefined trajectories while detecting information such as the end-effector's pose and joint driving forces. A total of 100 motion trajectories are selected, each corresponding to different characteristics such as step height and step length. Each trajectory starts from an initial point and passes through 200 sampling points with a sampling period of 10 seconds, resulting in a self-built robot dynamics dataset.

(2) State Variables

For the rehabilitation training robot, the end-effector has 6 DOFs. The displacement and velocity of the end-effector in the 6 DOFs are selected as state variables, forming a 12-dimensional state space. The state variables are defined as $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5, x_6, v_1, v_2, v_3, v_4, v_5, v_6]^T$. The robot has a total of 9 DOMs, with 6 active joints in the 6SSP device and 3 active joints in the 3RPS device. Therefore, the control variables are chosen as the joint driving torques for the 9 moving joints, defined as $\mathbf{\tau} = [\tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7, \tau_8, \tau_9]^T$.

(3) State Lifting Functions

The selection of Koopman state lifting functions considers several aspects. First, the lifting functions should capture the important dynamic characteristics of the system, with common choices including polynomial functions, Fourier basis functions, or radial basis functions. Second, the dimensionality of the

lifting functions should be appropriate; too low may fail to adequately describe the system dynamics, while too high may lead to increased computational complexity and overfitting risk. Finally, optimizing the combination of lifting functions can further enhance the model's effectiveness and performance. Through prior knowledge and simulation experiments, the state lifting function is determined to be a Thinplate radial basis function with a dimensionality of 100. This state lifting function maps the original 12-dimensional state vector into a 100-dimensional high-dimensional space.

(4) Establishment and Verification of the Koopman Dynamic Model

A Koopman approximate operator is constructed to describe the linear evolution of state variables. The least squares method is used to fit the Koopman operator, thus learning the linear mapping relationship. Ultimately, the Koopman approximate operator $\mathbf{K}$ is obtained, satisfying the equation $\mathbf{z}_{k+1} = \mathbf{K} \cdot \mathbf{z}_k$. Given the high-dimensional state $\mathbf{z}_k$, it evolves linearly through $\mathbf{K}$, continuously predicting the subsequent behavior of the system in high-dimensional space. In this study, random points are taken from the original data of the robot's motion trajectories. High-quality and abundant data can help construct a more accurate Koopman operator, thereby improving the prediction accuracy. The trained Koopman model is used to predict the subsequent five steps. The **Figure 4** illustrates a comparison between the true values of the dynamic model at a given sampling point and its subsequent five steps, and the predicted results obtained using EDMD with different basis dimensions (80, 100, and 120 dimensions).

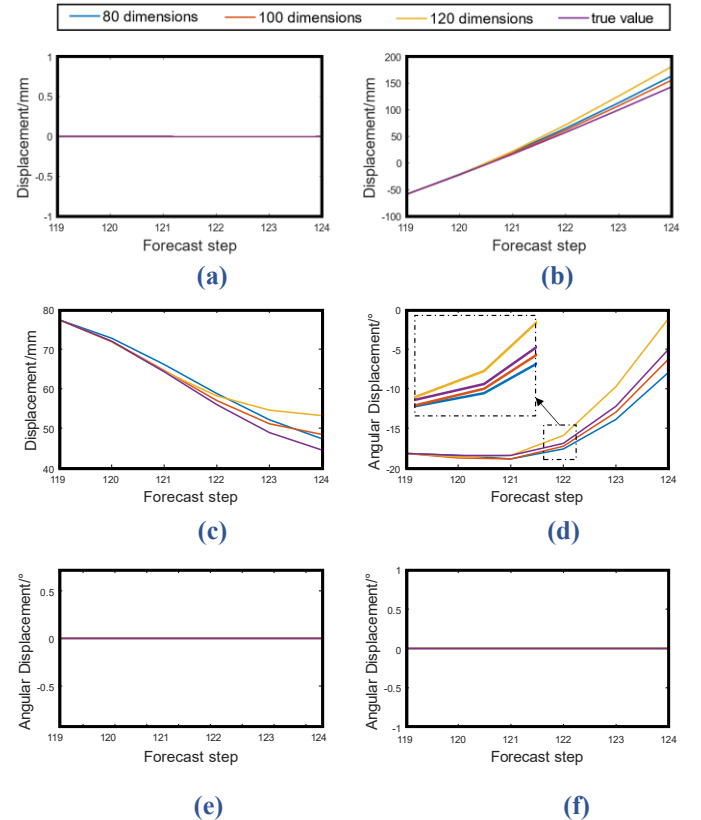


Figure 4. The recursive prediction results for 5 steps.(a) Displacement in the Y direction; (b) Displacement in the X direction; (c) Displacement in the Z direction; (d) Rotation angle around the Y direction; (e) Rotation angle around the X direction; (f) Rotation angle around the Y direction.

Table 1. The MSE of the predictions for the three Koopman dynamic models

	X	Y	Z	roll	patch	yaw
80 dimensions	0	194.17	8.28	2.71	0	0
100 dimensions	0	18.15	0.14	0.04	0	0
120 dimensions	0	1.36e+03	237.08	32.28	0	0

Table 1 illustrates the mean squared error (MSE) of the predictions over five steps for the three Koopman dynamic models. The results indicate that when the basis function dimension is 100, the Koopman model's predicted results are closest to the true values. This suggests that, compared to other dimensions such as 80 and 120, the 100-dimensional basis function is the optimal configuration for the Koopman dynamic model, achieving the highest prediction accuracy. Furthermore, the model demonstrates more precise predictive capabilities within the first two time steps. However, as the number of prediction steps increases, the cumulative error grows, leading to greater inaccuracies. For predictions in motion directions with stable states, the Koopman method can effectively extract features, resulting in a fitting error of zero. If the system exhibits highly nonlinear characteristics, the model may require more basis functions or complex dynamic modes to accurately capture the system behavior. High-quality and abundant data can aid in constructing a more precise Koopman operator, thereby enhancing the accuracy of predictions.

CONCLUSIONS

The advantages of the Koopman method in establishing robot dynamics models include its efficient handling of complex nonlinear systems, good interpretability, and enhanced system stability and predictive capabilities. By mapping nonlinear dynamics to a linear space, the Koopman method facilitates the application of linear control theory, enabling precise control and optimization. Additionally, this method supports real-time data updates, allowing it to adapt to dynamic environmental changes. In terms of application outcomes, the Koopman method has been successfully applied in areas such as path planning, autonomous navigation, and motion control, significantly improving the performance and safety of robots.

The Koopman operator method is data-driven, enabling the automatic extraction of the system's dynamical characteristics from the robot's motion and operational data, without relying on precise physical modeling. The Koopman operator constructs a high-dimensional state space that encapsulates the system's complexity within a data-driven linear model, thereby making the understanding and prediction of system behavior simpler and more accurate. This paper investigates the application of the

Koopman operator method in dynamics modeling for redundant rehabilitation robots. By utilizing a substantial amount of data related to robot motion and mechanical information, it was found that the Koopman approximate operator can effectively capture the linear evolution relationships of the system in high-dimensional space, thus predicting robotic dynamics behavior within that space and achieving precise predictions within a limited number of steps. However, the model still faces challenges when dealing with complex nonlinear systems and multi-step predictions. Improving the selection of function forms and dimensions is crucial for enhancing the model's accuracy. Future research should focus on further improving the model's accuracy in multi-step predictions to meet the increasingly complex and precise control demands.

FUNDINGS

This work was supported in part by national key R&D program (Ministry of science and technology of the People's Republic of China : 2019YFB1311400)

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